

Deadline and meetings for the tasks

1. 25th June (code Implementation – Samples- Presentation in codes and results)
2. 1st July (Documentation structure pipelines – what to write and include)
3. 3rd July (Final deadline to submit the documentation)

Agreed Tasks:

1. Preprocessing steps in order:

- a. Abhirami:
 - i. Take screen shots from the videos for each fish species and create a directory for each
- b. Jibin + Tuna:
 - i. Data Cleaning
 - ii. Data Preprocessing
 - iii. Data Augmentations
 - iv. Quality features (check if quality is ok)
- c. Parsa:
 - i. Image dehazing and clarity

2. Training:

- a. Jibin + Abhirami:
 - i. YoloV8
 - ii. Bot-sort
 - iii. Edge Weight Average
- b. Parsa
 - i. Sonar Audio Transmission (Convert to images and give images to preprocessing step) – If applicable
 - ii. DAT Training
 - iii. Optimization
 - iv. Boosting

3. Evaluation:

- a. Tuna
 - i. MAE
 - ii. MoTA
- b. Tuna + Jibin:
 - i. Some other required metrics w.r.t training data

- c. Abhirami:
 - i. Statistical Data Analysis
- d. Jibin:
 - i. Expected Calibration tracking
- e. Parsa:
 - i. Dehazing, training, Bot-SORT Tracking evaluations
 - ii. Charts and Graphs analysis

4. Documentation:

All of us will write or own each results documentation. For the end to combine all of them Abhirami would combine all sections. For documentation something is important that must be clear which pipeline flowchart image which will be done by Parsa. For the rest we talk after the implementations.

- a. Parsa:
 - i. Pipeline Flowchart Creation